ITDS 124 Mathematics of Computing

Chapter 2. Logic (ตรรกศาสตร์)

2.1 Propositions and Logical Operations

(ประพจน์และการดำเนินการทางตรรกะ)

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Statement (ข้อความ) or Proposition (ประพจน์)

- A statement or proposition is a declarative sentence that is either true or false but not both.

- Examples of statements
  - The earth is round.
  - $2 + 3 = 5$
  - The temperature on the surface of the planet Venus is $800°\mathrm{F}$.
  - The sun will come out tomorrow.

Not a statement or proposition

- Question
- Command
- Request
- Exclamatory
- Include unknown variables i.e. x, y
- Examples of not statements
  - Do you speak English? ← Question, not a statement
  - Take two aspirins. ← A command, not a statement
  - $3 - x = 5$ ← is a declarative sentence, but not a statement since it is true or false depending on the value of x.

Logical Connectives (ตัวดำเนินการเชื่อมทางตรรกศาสตร์) and Compound Statements (ข้อความประกอบ)

- In math, $x, y, z, \ldots$ denote variables that can be replaced by real numbers, and these variables can be combined with operations $+, -, \times, \div$.
- In logic, $p, q, r, \ldots$ denote propositional variables that can be replaced by statements.
  - p: The sun is shining today.
  - q: It is cold.
- Statements or propositional variables can be combined by logical connectives to obtain **compound statements**.
  - $p\ and\ q$: The sun is shining today and it is cold.
  - $p\ or\ q$: The sun is shining today or it is cold.

Negation (นิเสธ)

- If p is a statement, the negation of p is the statement $\boldsymbol{not\ p}$ or $\boldsymbol{\sim p}$.
- If p is true, $\sim p$ is false.
- If p is false, $\sim p$ is true.
- not is not a connective since it does not join two statements.
- $\sim p$ is not really a compound statement.
- not is a unary operation for the collection of statements and $\sim p$ is a statement if p is a statement.
- Examples of negation
  - $p : 2 + 3 > 1$ $\quad\quad$ $\sim p : 2 + 3 \ \leq 1.$
  - $q : It\ is\ cold$ $\quad\quad$ $\sim q : It\ is\ not\ cold$

p	$\sim q$
T	F
F	T

Conjunction (AND) (การเชื่อม)

- If p and q are statements, the conjunction of p and q is the compound statement "$\boldsymbol{p}$ ***and*** $\boldsymbol{q}$" denoted by $\boldsymbol{p} \wedge \boldsymbol{q}$.
- and is a binary operation on the set of statements.
- The truth table (ตารางค่าความจริง) shown below :

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction (OR) (การเลือก)

- If p and q are statements, the disjunction of p and q is the compound statement "***p or q***" denoted by $\boldsymbol{p} \vee \boldsymbol{q}$.
- $\boldsymbol{p} \vee \boldsymbol{q}$ is true when "either p or q may be true or they may both be true at the same time", sometimes we call "**inclusive or**", "**inclusive disjunction**"
- or is a binary operation on the set of statements.
- The truth table shown here:

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Exclusive Disjunction (XOR)

- If p and q are statements, the exclusive or of p and q is the compound statement "either p or q may be true, but both are not true at the same time", denoted by $\boldsymbol{p \oplus q}$.
- $exclusive\ or$ is a binary operation on the set of statements.
- The truth table shown here :

$\boldsymbol{p}$	$\boldsymbol{q}$	$\boldsymbol{p \oplus q}$
T	T	F
T	F	T
F	T	T
F	F	F

Exclusive versus Inclusive

- Depending on the circumstances, some disjunctions are inclusive and some are exclusive.
- Examples of Inclusive
  - “I have a dog” or “I have a cat”
  - “It is warm outside” or “It is raining”
- Examples of Exclusive
  - Today is either Tuesday or it is Thursday
  - Pat is either male or female

Compound statements (ข้อความประกอบ)

- A compound statement is a statement made from other statements by using logical connectives.
- s: $p \vee (q \wedge (p \vee r))$ involves three propositions, p, q, and r.
- There are $2^3 = 8$ possible combinations of truth values for p, q, and r.
- Truth table can be systematically constructed in the following way.
  - The first n columns of the table are labeled by the component propositional variables. Further columns are included for all intermediate combinations of the variables, culminating in a column for the full statement.
  - Under each of the first n headings, we list the 2^n possible n-tuples of truth values for the n component statements
  - For each of the remaining columns, we compute, in sequence, the remaining true values.

Example of making a truth table

- Make a truth table for the statement $(p \wedge q) \vee (\sim p)$.

p	q	$p \wedge q$	$\vee$	$\sim p$
T	T	T	T	F
T	F	F	F	F
F	T	F	T	T
F	F	F	T	T

Predicate

- An element of $\{x \mid \mathrm{P}(x)\}$ is an object t for which the statement $\mathrm{P}(t)$ is true.
- $\mathrm{P}(x)$ is called a predicate or a propositional function because each choice of x produces a proposition $\mathrm{P}(x)$ that is either true or false.
- Ex. Let $A = \{x \mid x \textit{ is an integer less than } 8\}$.
  - $\mathrm{P}(x)$ is the sentence "$x \textit{ is an integer less than } 8$".
  - $\mathrm{P}(1)$ is true since "$1 \textit{ is an integer less than } 8$", so $1 \in A$.

Universal quantifier (∀) (ตัวบ่งปริมาณสากล/ทั้งหมด)

- The universal quantification of a predicate $\mathrm{P}(x)$ is the statement "For all values of x, $\mathrm{P}(x)$ is true, and is denoted by $\forall x\ \mathrm{P}(x)$.
- Universal quantification → "for every x", "every x", or "for any x".
- The symbol ∀ is called the universal quantifier.
- Example
  - The sentence $\mathrm{P}(x): -(-x) = x$. $\forall x\ \mathrm{P}(x)$ is a true statement because for all real numbers, $-(-x) = x$.
  - Let $\mathrm{Q}(x): x + 1 < 4$. $\forall x\ \mathrm{Q}(x)$ is a false statement, because $\mathrm{Q}(5)$ is not true.
- The order in which multiple ∀ quantifications are considered does <u>not</u> affect the truth value. $\forall x \forall y\ \mathrm{P}(x, y) = \forall y \forall x\ \mathrm{P}(x, y)$

Existential quantification (∃) (ตัวบ่งปริมาณบางส่วน)

- The existential quantification of a predicate $\mathrm{P}(x)$ is the statement "There exists a value of x, $\mathrm{P}(x)$ is true, and is denoted by $\exists x\ \mathrm{P}(x)$.
- Existential quantification → "there is an x", "there is some x", or "there exists an x", "there is at least one x".
- The symbol ∃ is called the existential quantifier.
- Example
  - Let $\mathrm{Q}(x): x + 1 < 4$. $\exists x\ \mathrm{Q}(x)$ is a true statement, because $\mathrm{Q}(2)$ is true.
- The order in which multiple ∃ quantifications are considered does <u>not</u> affect the truth value. $\exists x \exists y\ \mathrm{P}(x, y) = \exists y \exists x\ \mathrm{P}(x, y)$

Precedence of Quantifiers (ลำดับความสำคัญ)

- The quantifiers ∀ and ∃ have higher precedence than all logical operators.
- Ex. $\forall x\, \mathrm{P}(x) \vee \mathrm{Q}(x)$ is the disjunction of $\forall x\, \mathrm{P}(x)$ and $\mathrm{Q}(x)$.
- Ex. $\forall x\, \mathrm{P}(x) \vee \mathrm{Q}(x)$ is the not the same as $\forall x\, (\mathrm{P}(x) \vee \mathrm{Q}(x))$.

Applying both universal & existential quantifications

- Order of application does matter
- Example : Let A and B be $n \times n$ matrices.
  - $\forall A\ \exists B\ \ A + B = I_n$ ← true
  - $\exists B\ \forall A\ \ A + B = I_n$ ← false
  - $\exists B\ \forall A\ \ A + B = A$ ← true

Applying negation to quantifiers

- Let p: $\forall x\ \mathrm{P}(x)$.
- The negation of p is false when p is true and true when p is false.
- For p to be false there must be at least one value of x for which $\mathrm{P}(x)$ is false. Thus, p is false if $\exists x \sim\mathrm{P}(x)$ is true.

$$\boldsymbol{\sim\forall x\ \mathrm{P}(x) = \exists x \sim\mathrm{P}(x)}$$

- If $\exists x \sim\mathrm{P}(x)$ is false, then for every x, $\sim\mathrm{P}(x)$ is false; that is $\forall x\ \mathrm{P}(x)$ is true.

$$\boldsymbol{\sim\exists x\ \mathrm{P}(x) = \forall x \sim\mathrm{P}(x)}$$

Applying negation to quantifiers

- Example 1:
  - Let p : For all positive integers $n = n^2 + 41n + 41$ is a prime number.

    $\sim p$: There is at least one positive integer n for which $n^2 + 41n + 41$ is not prime.
- Example 2:
  - Let q : There is some integer k for which $12 = 3k$.

    $\sim q$: For all integers $k, 12 \neq 3k$.